

# Thermodynamics

Lecture 4B:

- Energy analysis of closed systems

Prof. Claudio Hail (he/him)

# Learning objectives

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What you will be able to do after this course:

- I. *Knowledge:* Write the first and second laws of thermodynamics.**
- II. *Comprehension:* Describe the first and second laws in the student's own language.**
- III. *Application:* Solve simple single-answer problems using the first law.**
- IV. *Analysis:* Solve problems requiring both the first and second laws.
- V. *Synthesis:* Develop schemes to improve the efficiency or performance of thermodynamic systems, e.g., compressors, turbines, heat pumps, etc.

# Learning objectives (II)

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- VI. Evaluation: Determine and describe second law fallacies in proposed power cycles.
- VII. Evaluation: Judge when classical thermodynamics is not the appropriate analysis tool.
- VIII. Evaluation: Find and correct errors in the student's own solutions and in those of others.
- IX. *Analysis and Evaluation:* Search appropriate databases and the literature to find required thermodynamic data, and if the data are not available the student can select appropriate procedures and predict the values of the data.**
- X. Knowledge: Briefly outline the history of the field of thermodynamics.

# Recap Lecture 4A

## Boundary work

### Isochoric

$$W_b = \int_{V_1}^{V_2} p dV = 0$$

### Isobaric

$$W_b = p_0(V_2 - V_1)$$

### Isothermal

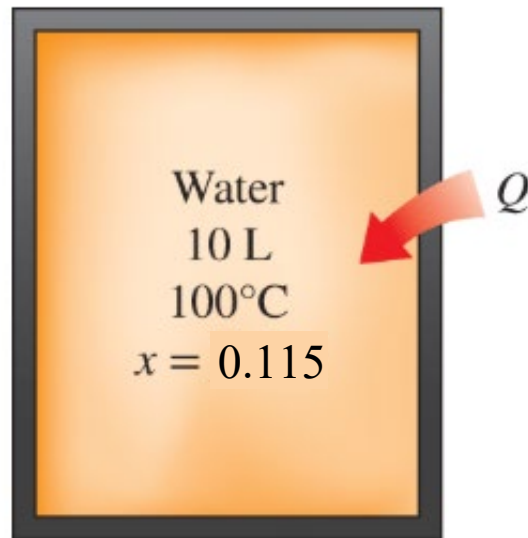
$$W_b = mRT_0 \ln \frac{V_2}{V_1}$$

### Polytropic

$$W_b = \int_{V_1}^{V_2} \frac{p_1 V_1^n}{V^n} dV = \frac{p_2 V_2 - p_1 V_1}{1 - n}$$

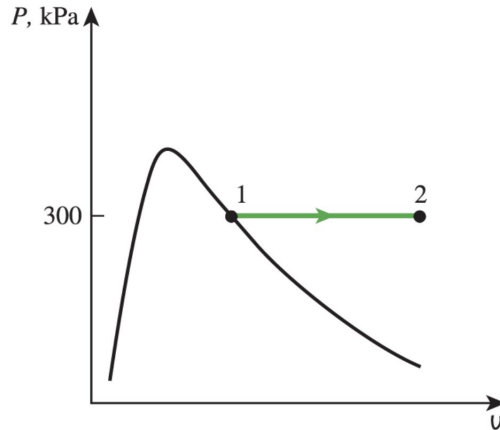
## General energy analysis

$$Q - W = \Delta U + \Delta KE + \Delta PE$$

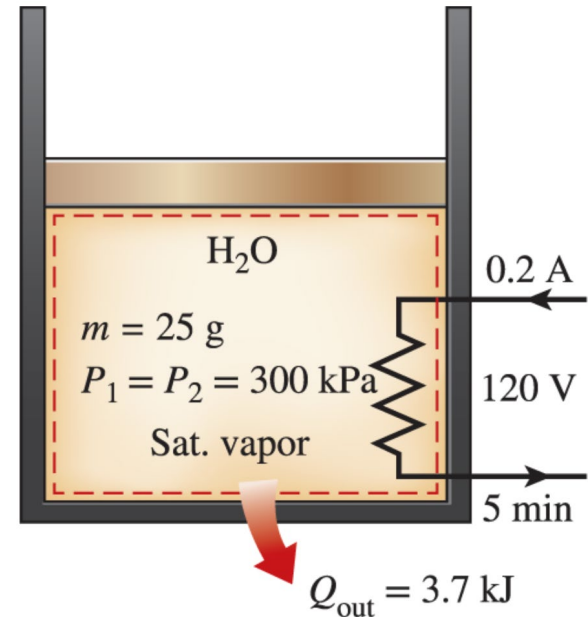


# Example II

- Constant pressure electrical heating of water



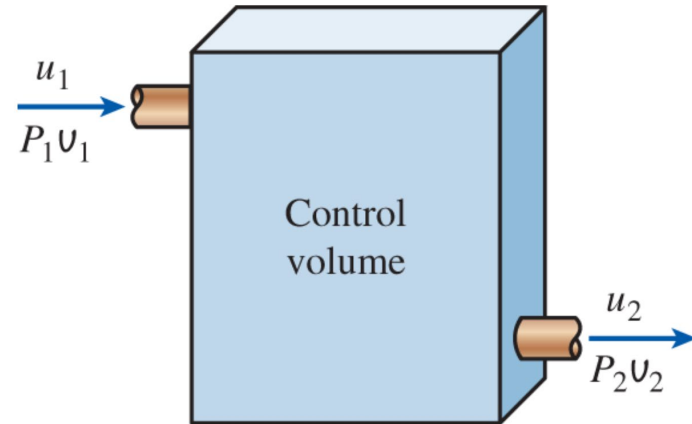
THIS EXAMPLE IS THE EXCEPTION OF THE ONLY CASE  
WHERE  $H$  IS USED FOR CLOSED SYSTEMS



# Enthalpy

- A combined property of internal energy and flow energy.
- Important in the analysis of open systems

$$h = u + Pv \left[ \frac{J}{kg} \right]$$



# After this lecture

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- Understand definitions and approximations for the internal energy, enthalpy, and specific heats of ideal gases, liquids, and solids.
- Learn how to perform a general energy analysis on a closed system for ideal gases, liquids and solids.

# Motivation

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A large gasketed plate heat exchanger

# Demonstration: Piston-Cylinder Compression

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- A cylinder is filled with ambient air and a small piece of cotton.
- Force is manually applied on the piston to compress the cylinder chamber volume.
- At sufficiently high temperatures cotton ( $>170^{\circ}\text{C}$ ) will ignite.



# Specific heat (I)

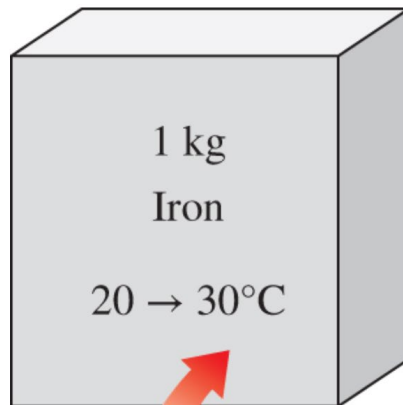
- Energy required to raise the temperature of a unit mass of substance by one degree.

Specific heat (capacity)

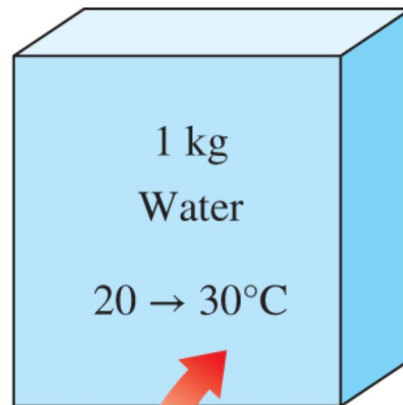
$$c = \left[ \frac{J}{kg \ K} \right]$$

Heat capacity

$$C = \lim_{\Delta T \rightarrow 0} \frac{\Delta Q}{\Delta T}$$



4.5 kJ



41.8 kJ

# Specific heat (II)

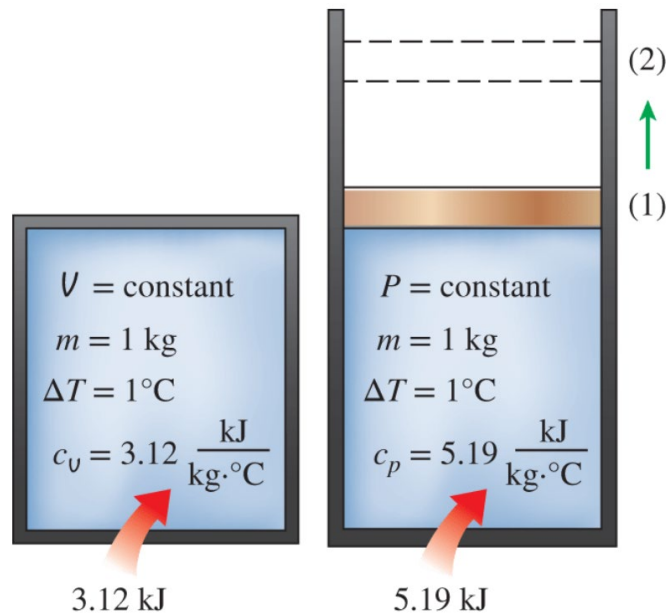
- Specific heats are thermodynamic properties obeying the state postulate.

Constant volume

$$c_v = \left( \frac{\partial u}{\partial T} \right)_v$$

Constant pressure

$$c_p = \left( \frac{\partial h}{\partial T} \right)_p$$



# Internal energy of ideal gases

- For ideal gases, the internal energy depends only on temperature

$$u = u(T)$$

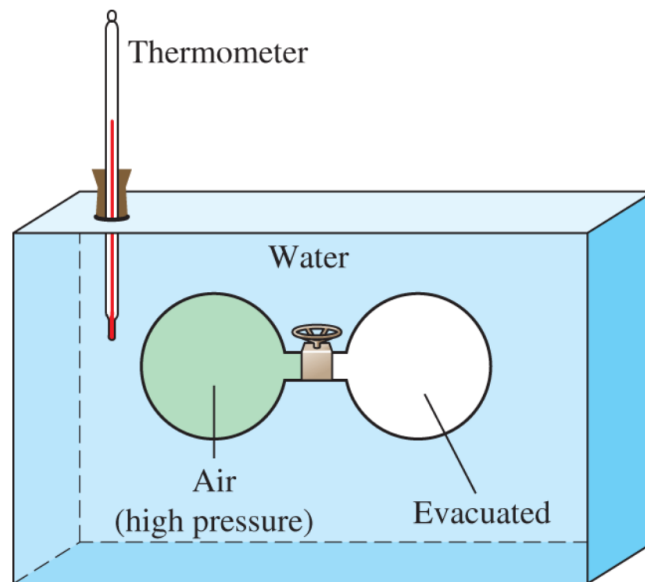
Experiment

$$\Delta U = Q - W$$

$$\Delta U = Q = 0$$

$$u_1 = u_2 \quad T_1 = T_2 \quad p_1 \neq p_2$$

See Book Chapter 12 for derivation



Joule's experimental set-up

# Enthalpy of ideal gases

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- Similarly, the enthalpy of ideal gases also only depends on temperature.

$$Pv = RT$$

$$u = u(T)$$

$$h = u(T) + pv = u(T) + RT = h(T)$$

Both  $u$  and  $h$  are tabulated in Appendix 1 ,Table A-17-25 for Air, N<sub>2</sub>, O<sub>2</sub>, CO<sub>2</sub>, CO, H<sub>2</sub>, H<sub>2</sub>O, O, OH

# Specific heats of ideal gases (I)

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- Partial derivatives become ordinary derivatives:

Change in internal energy:

$$c_v = \left( \frac{du}{dT} \right)_v$$

$$du = c_v(T) dT$$

$$\Delta u = u_2 - u_1 = \int_1^2 c_u(T) dT$$

Change in enthalpy:

$$c_p = \left( \frac{dh}{dT} \right)_p$$

$$dh = c_p(T) dT$$

$$\Delta h = h_2 - h_1 = \int_1^2 c_p(T) dT$$

Polynomial coefficients of  $c_p(T)$  are tabulated in Appendix 1 ,Table A-2c for many common gases

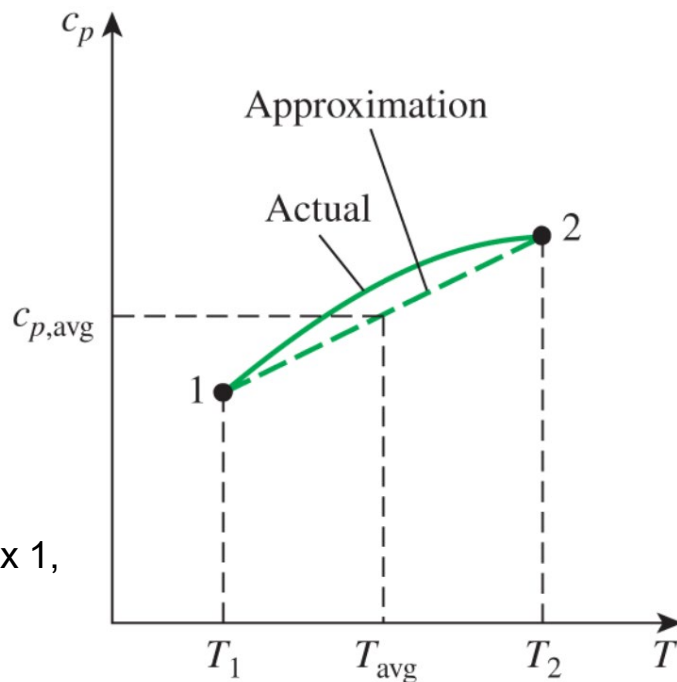
# Specific heats of ideal gases (II)

- Averaging specific heats over temperature allows for simpler calculation

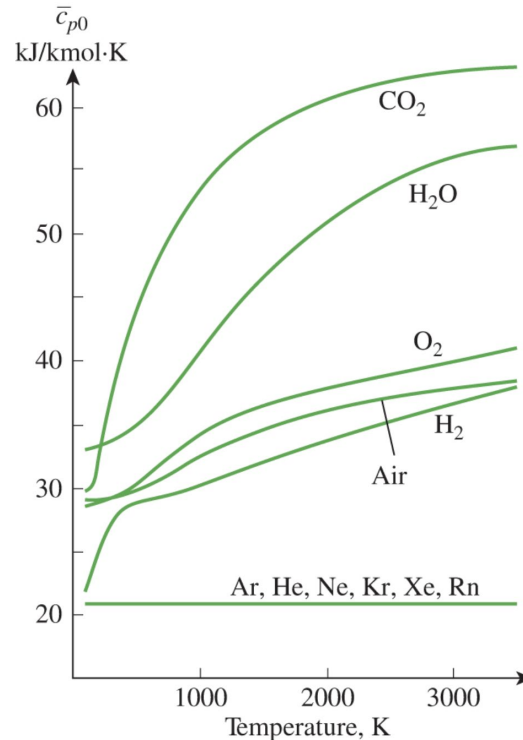
$$u_2 - u_1 = c_{v,avg}(T_2 - T_1)$$

$$h_2 - h_1 = c_{p,avg}(T_2 - T_1)$$

Tabulated values  $c_v(T)$  and  $c_p(T)$  are given in Appendix 1, Table A-2b for Air, CO<sub>2</sub>, N<sub>2</sub>, CO, H<sub>2</sub>, O<sub>2</sub>.



# Specific heats of ideal gases (II)



# Specific heat relations of ideal gases

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$$h = u(T) + RT$$

$$dh = du + RdT$$

Specific heat

$$c_p = c_v + R$$

Molar specific heat

$$\bar{c}_p = \bar{c}_v + \bar{R}$$

Specific heat ratio

$$k = \frac{c_p}{c_v}$$

# Demonstration: Piston-Cylinder Compression

How much work is required to ignite cotton?

Assume air is an ideal gas

$$R = 287 \text{ J/kgK} \quad c_v = 718 \text{ J/kgK}$$

State 1

$$V_1 = \pi r^2 L_1 = 1.17 \cdot 10^{-5} \text{ m}^3$$

$$p_1 = 1 \text{ atm}$$

$$T_1 = 20^\circ \text{C} = 293.15 \text{ K}$$

$$\text{First Law: } \Delta U = Q - W$$

$$W = -m(u_2 - u_1) = -mc_v(T_2 - T_1) = 1.53 \text{ J}$$

State 2

$$V_2 = \pi r^2 L_2$$

$$V_2 = 0.785 \cdot 10^{-6} \text{ m}^3$$

$$T_2 = 170^\circ \text{C}$$

$$m = \frac{V_1 p_1}{RT_1} = 14.2 \text{ mg}$$



State 1



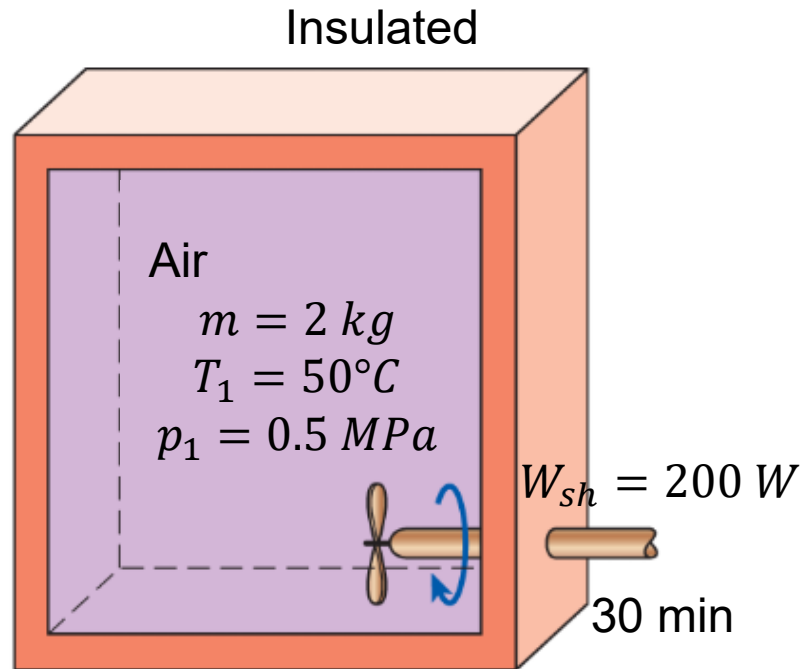
State 2

# Example: Ideal gas general energy analysis

A paddle wheel is operated in an insulated container for 30 min.  
Determine:

- P-V diagram of the process.
- The final temperature.
- The final pressure.

Assume  $c_p = c_p(50^\circ)$  and  $c_v = c_v(50^\circ)$  are constant over the considered temperature range.  
 $R = 278 \text{ J/kgK}$ .



# Group Work: in groups of 2; discuss the following question

## Given:

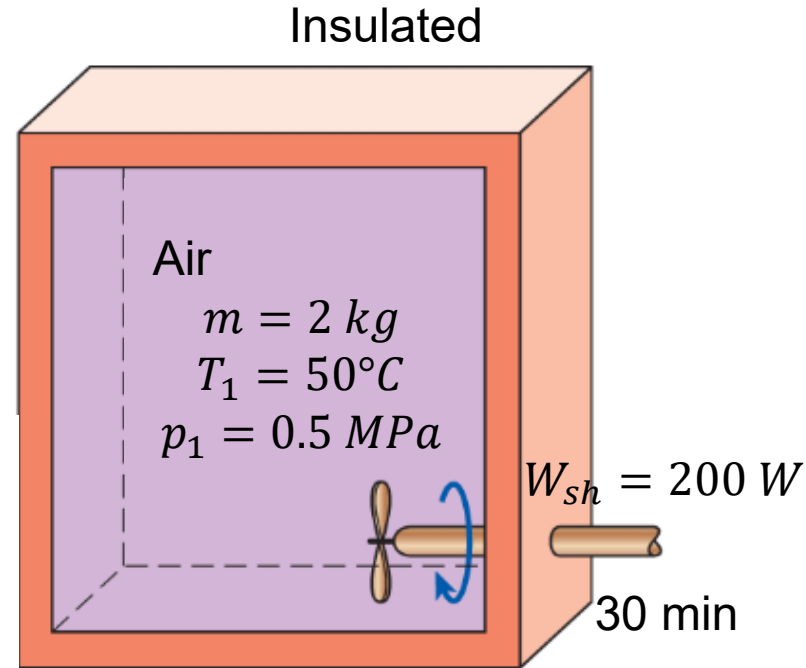
A paddle wheel is operated in an insulated container for 30 min.

## Determine:

1. P-V diagram of the process.
2. The final temperature.
3. The final pressure.

## To Do:

- Go to [pollev.com/claudiohail](https://pollev.com/claudiohail)
- Enter the first names of your group members, e.g. Olivia and Liam.
- Submit your answers to 1, 2 and 3.



# Group Work: in groups of 2; discuss the following question

| Temperature,<br>K | $c_p$<br>kJ/kg·K | $c_v$<br>kJ/kg·K | $k$ |
|-------------------|------------------|------------------|-----|
|-------------------|------------------|------------------|-----|

*Air*

|     |       |       |       |
|-----|-------|-------|-------|
| 250 | 1.003 | 0.716 | 1.401 |
| 300 | 1.005 | 0.718 | 1.400 |
| 350 | 1.008 | 0.721 | 1.398 |
| 400 | 1.013 | 0.726 | 1.395 |
| 450 | 1.020 | 0.733 | 1.391 |
| 500 | 1.029 | 0.742 | 1.387 |
| 550 | 1.040 | 0.753 | 1.381 |
| 600 | 1.051 | 0.764 | 1.376 |
| 650 | 1.063 | 0.776 | 1.370 |

## Given:

A paddle wheel is operated in an insulated container for 30 min.

## Determine:

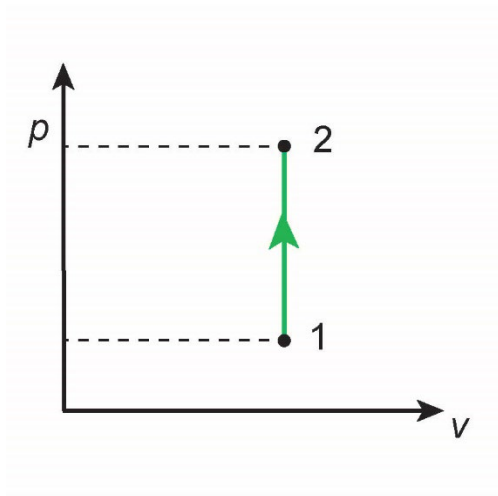
1. P-V diagram of the process.
2. The final temperature.
3. The final pressure.

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# Example: Ideal gas general energy analysis



First law:  $Q - W = \Delta U$

Shaft work  $W = -200 \text{ W} \cdot 30 \cdot 60 \text{ s} = -360 \text{ kJ}$

$$\Delta U = mc_{v,avg}(T_1 - T_2) = -W$$

Solved for  $T_2 = T_1 - \frac{W}{mc_{v,avg}}$

$$c_{v,avg} \approx c_v(T_1) \quad T_1 = 323.15 \text{ K} \quad m = 2 \text{ kg}$$

Interpolate  $c_v$  from table

$$c_v(T_1) = c_v(300\text{K}) + \frac{c_v(350\text{K}) - c_v(300\text{K})}{350\text{K} - 300\text{K}}(T_1 - 300\text{K}) = 0.7194 \frac{\text{kJ}}{\text{kgK}}$$

Plug in:  $T_2 = 573.36 \text{ K}$

# Summary

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- Understand how to describe different cases of work due to moving boundaries.
- Perform a general energy analysis on closed systems.

# Next up

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- General energy analysis of control volumes / open systems.

